

Solvency Field Theory V13.6

From Quantum Settlement Rules to Atomic Structure and Chemical Bonding

Preservation label: **SFT_V13_6_QUANTUM_TO_CHEMISTRY_MECHANISM_STACK_FROZEN**

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Recommended verdict

ROLLUP_READY_R275Q_V13_6_QUANTUM_TO_CHEMISTRY_MECHANISM_STACK_FROZEN_AS_PARTIAL_PASS_LOCALLY_REAL_SETTLEMENT_RULES_CONNECT_CURRENT_WRITE_UNCERTAINTY_TUNNELING_PAULI_EXCLUSION_SHELL_CAPACITY_AUFBAU_SKELETON_AND_FIRST_COVALENT_BOND_EARNED_KERNELS_INCLUDE_D_EFF_EQUALS_C_RSPIN_EQUALS_HBAR_OVER_2M_CLOSURE_NORMALIZED_UNCERTAINTY_ROUTE_TRANS_BARRIER_EVANESCENT_SETTLEMENT_INTENSITY_PAULI_ZERO_SAME_STATE_WRITE_HAZARD_SUBSHELL_CAPACITY_TWO_TIMES_TWOELL_PLUS_ONE_SHELL_CAPACITY_TWO_N_SQUARED_QUALITATIVE_N_PLUS_ELL_SETTLEMENT_TAX_AND_H2_SHARED_SETTLEMENT_BASIN_OPEN_BOUNDARIES_FULL_SETTLEMENT_COVARIANCE_NATIVE_KAPPA_EXACT_ATOMIC_HAMILTONIAN_HUND_EXCHANGE_CORRELATION_NUMERIC_H2_VALENCE_QFT_AND_FULL_CHEMISTRY

Core thesis

The same locally real settlement mechanics that produce uncertainty, tunneling, and exclusion also generate shell capacity, approximate periodic-table filling structure, and the first qualitative covalent bond.

Abstract

SFT V13.6 connects locally real write mechanics to the first structures of chemistry. The branch begins by translating the old V11.2 dent-field uncertainty into V13 current-write language: a particle writes according to the current total settlement geometry, while its present state already contains recent-history drift. The closure-normalized diffusion scale is:

$$D_{\text{eff}} = c_{\text{r_spin}} = \hbar/(2m).$$

This gives the uncertainty route a native mechanical scale.

Tunneling is then translated as far-side write completion from evanescent settlement support. A barrier is a real settlement/action-cost ridge. It does not delete support; it attenuates it:

$$I_{\text{far}} \sim \exp(-2 \int \kappa dx).$$

The V13.5 hazard law maps that intensity to rare tunneling completion.

Pauli exclusion follows from 4π fermion closure: exchange flips settlement sign, same-state amplitude cancels, and the duplicate write channel has zero intensity and zero hazard.

That Pauli kernel gives electron shell capacity:

$$\begin{aligned} N_{nl} &= 2(2l + 1) \\ N_n &= 2n^2. \end{aligned}$$

Aufbau ordering is then given a native qualitative route: electron settlement addresses fill by radial band cost plus angular barrier tax,

$$T_{nl} \sim n + l,$$

with lower n winning ties by stronger nuclear penetration.

Finally, H2 bonding is translated as shared settlement support: two hydrogen wells overlap, electron support concentrates in the between-proton basin, opposite spin closures allow two electrons in that spatial basin, and a bond forms when shared settlement gain exceeds repulsion and localization costs.

This is a partial-pass mechanism freeze, not a full quantitative chemistry completion.

1. Scope and claim boundary

V13.6 is a mechanism-stack freeze. It does not claim that SFT has completed all of quantum mechanics, all of atomic physics, or all of chemistry. It claims something narrower and stronger:

SFT_HAS_NATIVE_ROUTES_TO_UNCERTAINTY_TUNNELING_PAULI_SHELLS_AUFBAU_SKELETON_AND_H2_BONDING

The allowed claim is:

V13_6_EARNS_A_PARTIAL_PASS_QUANTUM_TO_CHEMISTRY_MECHANISM_STACK

The blocked claims are:

SFT_HAS_SOLVED_ALL_QUANTUM_MECHANICS
SFT_HAS_DERIVED_FULL_CHEMISTRY
SFT_HAS_NUMERIC_H2_BOND_ENERGY_AND_LENGTH
SFT_HAS_EXACT_PERIODIC_TABLE_CONFIGURATIONS
SFT_HAS_FULL_QFT

Paper-safe sentence:

V13.6 does not complete quantitative chemistry; it establishes the locally real mechanism chain that makes chemistry possible.

Freeze-safe sentence:

The mechanism skeleton is frozen; the quantitative operator program remains open.

2. Mechanism arc

The V13.6 arc is:

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current-write uncertainty
-> tunneling
-> Pauli exclusion
-> electron shell capacity
-> Aufbau skeleton
-> first covalent bond
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Each link uses the same underlying SFT vocabulary: settlement support, settlement intensity, write hazard, no-overwrite, closure, and current total geometry.

3. R270Q - Current-write uncertainty

Gate result

R270Q translated V11.2's retarded dent field into V13 current-write language.

V11.2 had:

$$\psi = \bar{\psi} + \Delta \psi_{\text{ret}}.$$

V13.6 writes:

$$\Sigma_{\text{total}}(x,t) = \Sigma_{\text{self}}(x,t) + \Sigma_{\text{controlled}}(x,t) + \Delta \Sigma_{\text{unresolved}}(x,t).$$

The current write is:

$$x_{(n+1)} = F(x_n, u_n, \Sigma_{\text{total}}(x_n, t_n)).$$

Unresolved current geometry creates deterministic-but-unresolvable kicks:

$$x_{(n+1)} = x_{(n+1)}^{\text{clean}} + \xi_n.$$

Current write plus recent history

The key correction was not to put uncertainty only in the past. The current write is determined by the current total geometry, but the present state already contains previous unresolved deviations.

Thus observed uncertainty is:

$$\text{observed uncertainty} = \text{current-write dent} + \text{recent-history drift}.$$

Best sentence:

The worldline records every dent, but uncertainty is what the next write cannot resolve before it lands.

Status

PARTIAL_PASS_CURRENT_WRITE_UNCERTAINTY_ROUTE / D_EFF_TARGET_OPEN

Open boundary: the full settlement covariance operator remains to be derived.

4. R270R - Effective diffusion constant

R270R fixed the closure-normalized uncertainty scale.

The spin/closure half-radius is:

$$r_{\text{spin}} = \hbar / (2mc).$$

The one-speed write rate is c , so:

$$D_{\text{eff}} = c r_{\text{spin}} = \hbar / (2m).$$

Equivalently:

$$D_{\text{eff}} = c \lambda_C / (4\pi).$$

This identifies the V11.2/Nelson diffusion target as the one-speed write rate multiplied by the half-Compton closure radius.

Best sentence:

The uncertainty diffusion constant is the one-speed write rate multiplied by the half-Compton closure radius.

Relation to uncertainty

Once the diffusion constant is fixed at:

$$D_{\text{eff}} = \hbar/(2m),$$

the Fisher/Cauchy route gives the uncertainty scale:

$$\Delta x \Delta p \geq \hbar/2.$$

Status

PARTIAL_PASS_D_EFF_CLOSURE_NORMALIZED / MICRO_COVARIANCE_OPEN

Open boundary: derive the kick covariance directly from the full settlement operator:

$$\langle x_i x_j \rangle = 2 D_{\text{eff}} \delta_{ij} \Delta t.$$

5. R272Q - Tunneling from trans-barrier settlement intensity

Static barrier first

R272Q separated the solid base mechanism from the overextended dynamic barrier picture. Tunneling must work through ideal static barriers; therefore literal temporary gaps cannot be the base law.

A barrier is a settlement/action-cost ridge:

$$V(x) > E.$$

In the forbidden region, support changes from oscillatory to evanescent:

$$d^2 \psi/dx^2 = \kappa^2 \psi.$$

The decaying support is:

$$\psi_{\text{far}} \sim \exp(-\int \kappa dx).$$

The far-side settlement intensity is:

$$I_{\text{far}} \sim \exp(-2 \int \kappa dx).$$

V13.5 hazard selection gives:

$$\lambda_{\text{far}} = \Gamma I_{\text{far}}.$$

Thus:

$$P_{\text{tunnel}} \sim \exp(-2 \int \kappa dx).$$

Best sentence:

Tunneling is rare far-side write completion from closure support that survives exponential taxation by the barrier's settlement ridge.

Dynamic barriers

Real barriers are made of moving particles. Their settlement geometry fluctuates:

$$\kappa(x,t) = \kappa_0(x) + \delta \kappa(x,t).$$

This can modulate tunneling, but it is not the base law.

Status

Open boundary: derive kappa from the native settlement/write action.

6. R271Q/R271R - Pauli exclusion kernel

Fermion closure and exchange

The 4π electron closure gives the spinor sign:

$$\begin{aligned} R(2\pi) \psi &= -\psi \\ R(4\pi) \psi &= +\psi. \end{aligned}$$

For two identical fermions, exchange carries the relative sign:

$$\Psi(1,2) = -\Psi(2,1).$$

Set both particles to the same complete state q :

$$\Psi(q,q) = -\Psi(q,q).$$

Therefore:

$$\Psi(q,q) = 0.$$

Settlement intensity vanishes:

$$I_{\text{same}} = |\Psi(q,q)|^2 = 0.$$

Using the R267E2 hazard law:

$$\lambda_{\text{same}} = \Gamma I_{\text{same}} = 0.$$

Best sentence:

Two fermions cannot occupy the same state because exchange flips settlement sign, so the same-address amplitude cancels to zero and no next-write channel exists.

A/B audit

The 5.5 conservative route and Opus strong closure route were separated.

The conservative route says: Pauli follows if exchange of identical 4π fermions equals a relative 2π closure rotation.

The strong route says: 4π closure itself is the physical reason exchange antisymmetry appears.

Adjudication:

PAULI_KERNEL_EARNED_CONDITIONALLY
FULL_SPIN_STATISTICS_THEOREM_OPEN

Boson contrast

For 2π closure:

$$\Phi(1,2) = +\Phi(2,1).$$

Same-state occupation does not cancel. Bosons can share.

Status

PARTIAL_PASS_PAULI_KERNEL_CONFIRMED_BY_AB_AUDIT / FULL_SPIN_STATISTICS_THEOREM_OPEN

7. R273A - Electron shell filling and periodic table skeleton

Complete settlement address

For atomic electrons, use the central-potential correspondence label:

$q = (n, l, m_l, m_s).$

SFT translation:

n = radial settlement band / action-node class
 l = angular closure / orbital support class
 m_l = orientation address
 m_s = spin/closure branch

Pauli says no two electrons may share the same complete q .

Subshell capacity

For fixed n, l , the orientation labels are:

$m_l = -l, \dots, +l.$

There are:

$2l + 1$

orientation addresses. Each has two spin/closure branches:

$m_s = +1/2, -1/2.$

Therefore:

$N_{nl} = 2(2l + 1).$

Examples:

s: $l=0 \rightarrow 2$
p: $l=1 \rightarrow 6$
d: $l=2 \rightarrow 10$
f: $l=3 \rightarrow 14$

Shell capacity

For shell n :

$l = 0, 1, \dots, n-1.$

Then:

$N_n = \sum_{l=0}^{n-1} 2(2l+1) = 2n^2.$

Shell capacities:

$n=1: 2$
 $n=2: 8$
 $n=3: 18$
 $n=4: 32$

Best sentence:

Electron shells are settlement-address classes, and Pauli says no two 4π electrons may occupy the same complete address.

Status

PARTIAL_PASS_SHELL_CAPACITY_SKELETON / FULL_ATOMIC_ENERGY_ORDERING_OPEN

8. R273B - Native Aufbau / settlement energy order

R273B gave a native qualitative route to the Aufbau/Madelung skeleton.

Effective settlement tax:

$T_{nl} \sim n + l.$

where:

n = radial settlement-band cost
l = angular settlement barrier / penetration tax

At equal n+l, lower n wins because lower-n support penetrates deeper into the nuclear settlement well.

(n+l) tie -> lower n favored.

This recovers the filling skeleton:

1s
2s
2p
3s
3p
4s
3d
4p
5s
4d
5p
6s
4f
5d
6p
7s
5f
6d
7p

Best sentence:

Aufbau order is the order in which electron settlement addresses minimize radial band cost plus angular barrier tax, with lower-n penetration winning ties.

Guardrail

This is not exact spectroscopy. It does not derive all real configurations, Hund's rules, exchange/correlation corrections, spin-orbit effects, or transition-metal and lanthanide anomalies.

Status

PARTIAL_PASS_QUALITATIVE_AUFBAU_ROUTE / EXACT_ATOMIC_HAMILTONIAN_OPEN

9. R274A - First chemical bond: H2 shared settlement support

Bonding basin

Two separated hydrogen atoms have two separated proton settlement wells and two separated electron supports. When the atoms approach, the electron supports overlap.

Bonding support:

$$\psi_{\text{plus}} \sim \psi_A + \psi_B.$$

The intensity between the protons is:

$$\begin{aligned} I_{\text{between}} &= |\psi_A + \psi_B|^2 \\ &= |\psi_A|^2 + |\psi_B|^2 + 2 \operatorname{Re}(\psi_A^* \psi_B). \end{aligned}$$

The cross term is the settlement-overlap gain.

Best sentence:

A covalent bond is a shared settlement basin where electron support between two nuclei lowers the total write cost more than the separated atomic wells.

Pauli-compatible pair

Two electrons can share the same spatial bonding basin only because they have opposite spin/closure labels:

$m_s = +1/2, -1/2.$

Thus they are not duplicate complete settlement addresses.

Antibonding

Antibonding support:

$\psi_{\text{minus}} \sim \psi_A - \psi_B.$

This produces a node between the nuclei. SFT translation:

bonding = settlement bridge
antibonding = settlement cut

Energy ledger

A bond forms if:

$E_{H2}(R) < 2 E_H.$

SFT ledger:

$\Delta E(R) = - \text{shared_gain}(R)$
+ proton_proton_repulsion(R)
+ electron_electron_repulsion(R)
+ localization_tax(R)
+ node_pairing_cost(R).

Bonding occurs when:

$\Delta E(R) < 0.$

Equilibrium occurs when:

$dE/dR = 0$
and
 $d^2E/dR^2 > 0.$

Status

PARTIAL_PASS_H2_COVALENT_KERNEL / NUMERIC_MOLECULAR_CHEMISTRY_OPEN

Open boundary: no exact H2 bond energy, bond length, molecular spectra, or full valence chemistry yet.

10. Integrated mechanism map

One-speed closure support

- > current write follows total settlement geometry
- > unresolved current geometry plus recent drift gives uncertainty
- > closure radius gives $D_{\text{eff}} = \hbar/(2m)$
- > barriers tax support exponentially
- > far-side intensity gives tunneling
- > 4π closure gives exchange antisymmetry
- > same-state intensity vanishes for identical fermions
- > Pauli creates finite electron address capacity
- > atomic addresses form shells and subshells
- > radial + angular settlement tax orders filling
- > overlapping atomic supports form shared bonding basins
- > first covalent bond appears

That is the V13.6 story.

11. Gate ledger

Gate	Short status	Earned kernel	Main open boundary
R270Q	PARTIAL_PASS	current geometry + recent-history drift	exact D_{eff} route

Gate	Short status	Earned kernel	Main open boundary
R270R	PARTIAL_PASS	$D_{\text{eff}} = c \, r_{\text{spin}} = \hbar / (2m)$	micro covariance
R271Q/R271R	PARTIAL_PASS	Pauli zero same-state write hazard	full spin-statistics theorem
R272Q	PARTIAL_PASS	tunneling from evanescent trans-barrier intensity	native kappa
R273A	PARTIAL_PASS	$2(2l+1), 2n^2$ shell capacity	energy ordering
R273B	PARTIAL_PASS	n+l settlement tax	exact atomic Hamiltonian
R274A	PARTIAL_PASS	H2 shared settlement basin	numeric molecular chemistry

12. Earned kernels

CURRENT_WRITE_UNCERTAINTY_ROUTE
 D_EFF_CLOSURE_NORMALIZED_HBAR_OVER_2M
 STATIC_TUNNELING_KERNEL
 PAULI_EXCLUSION_KERNEL
 SHELL_CAPACITY_SKELETON
 QUALITATIVE_AUFBAU_ROUTE
 FIRST_H2_COVALENT_BOND_KERNEL

13. Open boundaries

Quantum foundations

FULL_SETTLEMENT_COVARIANCE_OPERATOR
 NATIVE_KAPPA_FROM_FULL_ACTION
 FULL_RELATIVISTIC_SPIN_STATISTICS_THEOREM
 FULL_HILBERT_RECONSTRUCTION
 FULL_QFT_CORRESPONDENCE
 DETECTOR_AMPLIFICATION_AND_RECORD_OBJECTIVITY

Atomic physics

NATIVE_ATOMIC_HAMILTONIAN
 NUMERIC_ORBITAL_ENERGIES
 SCREENING_CONSTANTS
 HUND_RULES
 EXCHANGE_CORRELATION_NUMERICS
 SPIN_ORBIT_FINE_STRUCTURE
 CONFIGURATION_ANOMALIES

Chemistry

NUMERIC_H2_BOND_ENERGY_AND_LENGTH
 VALENCE_RULES
 HYBRIDIZATION
 IONIC_BONDING
 METALLIC_BONDING
 HYDROGEN_BONDING
 VAN_DER_WAALS_BONDING
 MOLECULAR_SPECTRA

14. Native glossary

Settlement support-readiness

The distributed readiness of possible next writes. Correspondence role: wavefunction support.

Settlement intensity

Magnitude-squared support measure controlling write hazard:

$$I_i = |\psi_i|^2 \\ \lambda_i = \Gamma I_i.$$

Current-write uncertainty

Unpredictability from unresolved current total settlement geometry plus recent-history drift.

Closure radius

For spin-1/2 massive support:

$$r_{\text{spin}} = \hbar/(2mc).$$

Tunneling

Rare far-side write completion from evanescent support that survives a barrier/action-cost ridge.

Pauli exclusion

Zero duplicate fermion write channel due to 4π exchange antisymmetry and same-state amplitude cancellation.

Settlement address

A complete state label/admissible write class, e.g.

$$q = (n, l, m_l, m_s).$$

Aufbau settlement tax

Qualitative ordering rule:

$$T_{nl} \sim n + l.$$

Shared settlement basin

Covalent bonding region where electron support stabilizes more than one nuclear well.

15. Paper outline

Proposed paper title:

Solvency Field Theory V13.6: From Quantum Settlement Rules to Atomic Structure and Chemical Bonding

Suggested structure:

1. Scope and claim boundary.
2. Current-write uncertainty.
3. Closure-normalized diffusion constant.
4. Tunneling.
5. Pauli exclusion.
6. Electron shell capacity.
7. Aufbau skeleton.
8. First chemical bond.
9. Open gates.
10. Conclusion.

16. Final preservation statement

SFT V13.6 freezes the quantum-to-chemistry mechanism stack: locally real settlement mechanics provide native routes for uncertainty, tunneling, Pauli exclusion, shell capacity, qualitative Aufbau ordering, and the first H₂ covalent bond, with exact operator derivations, spectroscopy, correlation, valence, and full chemistry left as open quantitative gates.